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Seasonal variations of menstrual cycle length in a large, US-based, digital cohort

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Introduction

Menstrual characteristics, including menstrual cycle length and menstrual regularity, have been recognized as important vital signs in people who menstruate¹. Overall, short menstrual cycles (e.g., cycles shorter than 26 days) were suggested to be associated with lower fecundity and higher risk of anovulation while evidence for long cycles (e.g., cycles longer than 29 days) with fecundity was mixed^{2–5}. Throughout the menstrual cycle, ovulation and endometrial development are dependent on hypothalamus-pituitary-ovary regulation and hormonal signaling. Additional factors that may impact menstrual cycle variation include circadian rhythms and light exposure^{6–8}.

Seasonal trends in human reproduction have been documented. Previous studies have shown seasonality in birth and fecundability in various countries and regions and demonstrated differential seasonal trends in reproductive outcomes by latitude^{9–11}. However, seasonality in fecundability was not consistent across studies. For example, one study of seasonality in fecundability reported modest seasonal variation in fecundability, with a peak in the late fall and early winter in participants from North America (across all states in US and provinces in Canada) and Denmark⁹. Other studies reported a peak in births between August and September in southern US and between March and April in Northern Europe,

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which corresponds to conception peaks between November and December in USA and between July and August in Northern Europe^{11–14}. Other studies also reported suggestive seasonal trends in reproductive hormones such as follicle stimulating hormones and prolactin^{15–18}. However, evidence for seasonal trends in menstrual cycle length were limited and inconsistent, potentially due to different study designs or variability in geographic, demographic, and climate characteristics of study location. If menstrual cycle length varies by season, it would provide support for the observed seasonal trends in fertility. Moreover, it could provide evidence for an environmental impact on menstrual cycles, which may be important in relationship climate change.

In this study, we used menstrual cycle data collected from cycle tracking smartphone applications among participants in the Apple Women's Health Study (AWHS), a large, nationwide, digital cohort, to examine and quantify seasonal patterns in menstrual cycle length. In addition, we considered possible differences in the seasonality of menstrual cycle length by age, latitude, and history of polycystic ovary syndrome (PCOS). We considered modifications by these factors because age and PCOS affects menstrual cycle length and variability, and seasonal patterns could differ by latitude.

Methods

Study design and population

The Apple Women's Health Study is a prospective digital cohort study¹⁹. Participants have been able to enroll in this study through the Apple Research app beginning in November 2019 until present. To be eligible for this study, participants must live in the US, be age 18 or older (19 in Alabama and Nebraska, and 21 in Puerto Rico), have menstruated at least once, be able to communicate in English, be the sole user of their iCloud account and iPhone, and provide written informed consent of participation at enrollment. For this analysis, we included eligible AWHS participants under age 50 who did not report menopause, hysterectomy, or oophorectomy at baseline, who enrolled in this study between June 1, 2020 and June 30, 2022, and who contributed at least one completed menstrual cycle after enrollment. We excluded participants above age 50 because menstrual cycle length was highly irregular in this age group²⁰. This study has been approved by the Institutional Review Board at Advarra (CIRB #PRO00037562) and has been registered in [Clinicaltrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT04196595) (NCT04196595).

Detailed information on the study design has been previously described¹⁹. Data collection was conducted using mobile phone applications ("apps"). Information on demographics, health conditions and behaviors, and menstrual status was collected through surveys in the Apple Research app at enrollment and was updated every 12 months after enrollment. Reproductive history information was collected once at baseline. Factors related to menstrual cycles, including hormone use, pregnancy, lactation, and menopause as well as major changes in physical and mental health, were collected in monthly surveys.

Exposure and Outcome Assessment

Participants can track their menstrual periods using the Cycle Tracking feature in the Apple Health app or other third-party apps that have participants' permission to share data with the Health app. We collected menstrual flow entries prospectively after enrollment. A menstrual cycle was defined as one or more consecutive days with tracked menstrual flow followed by at least 2 days of no tracked flow, and the first day of having menstrual flow was identified as the first day of the cycle, as defined previously²⁰. Cycles shorter than 10 days or longer than 90 days were excluded from the analysis as they were more likely to be tracking errors than natural menstrual cycles. For the remaining cycles, we excluded cycles that were atypically long and likely artifacts due to gaps in record-keeping using participant-specific thresholds as described in our previous study²⁰. Because information on factors related to menstrual cycles from the monthly surveys was only recorded after enrollment, we considered prospectively tracked cycles for this analysis. To minimize the impact of hormone use (including hormonal birth control pills, intrauterine device, patch, implant, shot, and vaginal ring), pregnancy, and lactation, we only included cycles that have been confirmed with having none of these events in the corresponding monthly surveys. We used the calendar month of the first day the cycle as the indicator of season.

Covariates

We considered demographical (e.g., age, race and ethnicity, socioeconomic status, and body mass index [BMI]) and behavioral (e.g., exercise, sleep disruption, stress, smoking, alcohol use, electronic nicotine, and marijuana) variables that could vary over time and/or have previously been linked with menstrual cycle length in the literature^{20–24}. Age at each menstrual cycle was calculated using the self-reported birth year at enrollment and the year of the first day of each cycle. Race and ethnicity were self-reported at enrollment with the following categories: White, non-Hispanic; Black or African American or African; Asian; Hispanic, Latino, Spanish and/or other Hispanic; American Indian or Alaska Native; Middle Eastern or North African; Native Hawaiian or Pacific Islander; and an option of none of these categories can fully describe me. Because the number of participants who identified themselves as American Indian or Alaska Native, Middle Eastern or North African, Native Hawaiian or Pacific Islander, or who indicated none of these categories can fully describe them were small, we combined these categories into one group while those who chose more than one category were combined into a separate group. Information on socioeconomic status and BMI was collected at enrollment and updated every 12 months. Socioeconomic status was measured by the highest education level and the MacArthur Scale of Subjective Social Status²⁵. This scale is a self-rated rank on a 'social ladder' from 0 (the lowest) to 9 (the highest) relative to others. BMI was calculated using self-reported height and weight. Information on behavioral factors was collected from monthly surveys, including changes in exercise and weight (decrease, increase, and no change), alcohol use (none, 1-5 times/month, 6-10 times/month, 11-15 times/month, and more than 16 times/month), binary indicators of major health changes (i.e., having experienced surgery, illness, or change of medication), major stressful events, sleep disruption, smoking, electronic nicotine use, and marijuana use. Covariates measured for each calendar month were linked to menstrual cycles started in the same month. We matched socioeconomic status and BMI, updated every 12 months, to menstrual cycles using the most recent measures prior to the start of the cycle.

Missing values in the covariates were treated using missing indicators, which affects 0.2-4% of the included menstrual cycles.

Statistical analysis

We used linear mixed effect (LME) models with random intercepts by participants to estimate the seasonal differences in menstrual cycle length and 95% confidence intervals. To evaluate possible residual confounding and overadjustment of these covariates, we fitted three separate models: 1) a crude model only adjusted for calendar year, 2) a parsimonious model additionally adjusted for age, race and ethnicity, BMI, and socioeconomic status, and 3) a full model additionally adjusted for all other covariates (changes in exercise and weight, alcohol use, major health change, stressful events, sleep disruption, smoking, electronic nicotine use, and marijuana use).

We first used a sinusoidal model for seasonality according to previous studies^{9,26}. More specifically, we first converted the calendar month of each cycle (ranging from 1 to 12/12) into an angle (e.g., March would be 3/12 or 1/4). Then, we fitted an LME model including a sine and a cosine function of that angle. This model estimates the months with the peak and trough of cycle length and the magnitude of the difference in cycle length between them. Angles corresponding to the timing of the peak and trough were estimated as the $\arctan(\beta_{\text{sine}}/\beta_{\text{cosine}})$ where β_{sine} and β_{cosine} are the coefficients of the sine and cosine terms in LME model. These angles were then converted back to month of the year⁹. Cycle length difference between the peak and trough was calculated as $2 \times \sqrt{\beta_{\text{sine}}^2 + \beta_{\text{cosine}}^2}$ ²⁶.

Confidence intervals of the estimated cycle length difference were estimated using bootstrap with 1000 iterations. The estimated cycle length difference has the smallest possible value of 0, which can lead to a false positive bias even if there is no seasonal variation in cycle length⁹. Therefore, we simulated the null distribution by first assigning the first cycle of each person to start on a random day of the year and calculating new start dates for the following cycles with the actual cycle length. Then we fitted the LME model with this new data to estimate the cycle length difference under the null hypothesis. This procedure was repeated 1000 times, and we reported the mean and the 5th and 95th percentiles of the cycle length differences to represent the null distribution.

The sinusoidal model assumes a single peak and trough that are 6 months apart over the 12-month duration, which may not hold true. Therefore, we fitted LME models with a categorical variable of calendar month for seasonality, adjusting for the same sets of covariates. We also considered a conditional linear model to estimate cycle length differences by calendar months within an individual by including a categorical variable of participant IDs, which can effectively control for unmeasured, time-invariant confounding.

In addition to the continuous measure of menstrual cycle length, we included a secondary outcome of experiencing a long menstrual cycle, which was defined as a cycle longer than 38 days according to the International Federation of Gynecology and Obstetrics²⁷. We then fitted a generalized linear mixed effect model with a binomial distribution and a conditional logistic regression model (corresponding to the linear mixed effect models and conditional linear models) to estimate the odds ratios and 95% confidence intervals of experiencing long

menstrual cycles by calendar months, adjusting for the same set of covariates as the fully adjusted linear models.

We also considered several sensitivity analyses. To further reduce the impacts of these events on menstruation, we restricted the analyses to cycles with no hormone use, pregnancy, and/or lactation in the previous 3 cycles. To further account for changes of menstrual cycles related to COVID-19 vaccination, we additionally controlled for receiving a vaccine dose in a cycle. Information on COVID-19 vaccination, including vaccination status and date(s) of receiving a vaccine dose, was collected from a COVID-19 Vaccine Update Survey²⁸. Cycles started at the end of a calendar may be more likely to be affected by exposures in the next month, therefore, we reassigned exposures measured in the next calendar month for cycles starting within the last 7 days of the month. To evaluate impacts of extremely long cycles, we restricted to cycles between the 5th and 95th percentiles of the cycle length distribution. Other sensitivity analyses included restricting to participants who contributed at least 6 cycles to the data and to cycles with confirmed accurate tracking. Tracking accuracy was identified when the participants reported ‘yes, they were accurate’ to the question ‘Are all your period days during the previous calendar month accurately reflected in the Health app?’ in the corresponding monthly survey. We also excluded menstrual cycles during which the participant enrolled in this study to minimize the impact of length-biased sampling²⁹.

To further explore possible effect modification, we fitted separate models by age (under 25, 25-29, 30-34, 35-39, and above 40), latitude (below 35°N, 35°-39°N, and above 40°N), and diagnosis of PCOS. Latitude was measured using the centroid of the state in which the participant resided as reported in the demographic survey and the cutoffs were determined based on a previous study to increase comparability between studies⁹. Diagnosis of PCOS was identified from self-reports in the Medical History survey at enrollment.

Data management, processing, and statistical analyses were conducted in R (version 3.6.0) and Python (version 3.6). All statistical tests were two-sided.

Results

A total of 125,104 menstrual cycles from 17,427 participants were included in the analysis (Figure S1). The median (interquartile range) of menstrual cycle length was 28 (26, 30) days. Distributions of demographic and lifestyle factors were presented on Table 1 and Table S1. On average, each participant contributed 7 cycles and the median was 5 cycles (interquartile range=2, 10). The mean age at enrollment was 33 years (SD=8) and more than 70% of the participants self-identified as White, non-Hispanic. More than 60% of the participants had at least a college degree (i.e., 4 year college or graduate school and above). All participants reported living in the northern hemisphere.

The sinusoidal model suggested menstrual cycles starting in April had the longest length and those starting in October had the shortest length (Table 2). However, the difference between the highest (i.e., the longest cycle) and the lowest was only 0.16 days (95%CI: 0.06, 0.26), which was larger than the 97.5th percentile of the permuted null distribution of cycle length

difference (0.11 days). Estimates from the crude, parsimonious, and fully adjusted models were similar.

Estimates for categorical variables of calendar month from the linear mixed effect models and conditional linear models were comparable (Figure 1), with the shortest menstrual cycle found in October. In addition, both the linear mixed effect models and the conditional linear models suggested a three-season pattern in changes of menstrual cycle length throughout the year. For example, there are no notable differences in cycle length among menstrual cycles starting from January to April. Meanwhile, menstrual cycles starting between January to April were relatively longer compared to cycles starting in other calendar months. Menstrual cycles starting in summer months from May to August were approximately 0.1-0.2 days shorter compared to January cycles (linear mixed effect model cycle length difference = -0.16, 95%CI: -0.28, -0.03 in May; -0.16, 95%CI: -0.29, -0.04 in June; -0.15, 95%CI: -0.29, 0.00 in July; and -0.08, 95%CI: -0.22, 0.06 in August; conditional linear models = -0.13, 95%CI: -0.26, -0.013 in May; -0.14, 95%CI: -0.27, -0.02 in June; -0.13, 95%CI: -0.28, 0.01 in July; and -0.09, 95%CI: -0.23, 0.06 in August). Cycles starting in September to December were even shorter compared to those starting in January by 0.2-0.3 days, with October having the shortest menstrual cycle length by 0.31 (95%CI = 0.17, 0.45) days in the linear mixed models and by 0.28 (95%CI= 0.14, 0.42) days in the conditional linear models, compared to January cycles. Results on experiencing long menstrual cycle (i.e., having a cycle > 38 days) supported the observed cycle length differences (Table S2).

All sensitivity analyses showed similar estimates to the main results (Table S3 and S4). Analyses restricting to cycles between 22-39 days and excluding enrollment cycles showed consistent trends for cycle length change across calendar month but with attenuated estimates (Table S3 and S4).

When stratified by age, the previously described seasonal patterns in menstrual cycle length were limited to participants under age 40. Moreover, those under age 30 had slightly larger seasonal changes in cycle length (cycle length difference comparing October vs January = -0.72, 95%CI=-1.16, -0.28 among those under age 25; =-0.47, 95%CI=-0.86, -0.08 among those between age 25-29. Estimates were from linear mixed effect models, same below). No seasonal patterns in cycle length were found by calendar month for participants above age 40 (Table 3 and Table S5). Analysis by latitude suggested stronger seasonal trends in menstrual cycle lengths among those who resided in states below 35°N (cycle length difference comparing October vs January = -0.23, 95%CI=-0.53, 0.06) and above 40°N (cycle length difference comparing October vs January=-0.48, 95%CI=-0.70, -0.26), compared to those from states between 35°-39°N (cycle length difference comparing October vs January=-0.09, 95%CI=-0.33, 0.14) (Table 4 and Table S6). Participants who reported a history of PCOS also showed stronger seasonal variation (cycle length difference comparing October vs January=-0.66, 95%CI=-1.25, -0.07) compared to those who did not report having been diagnosed with PCOS (cycle length difference comparing October vs January=-0.23, 95%CI=-0.37, -0.08) in both the linear mixed effect models and conditional linear models (Table 4 and Table S6).

Discussion

In the Apple Women's Health Study participants, we found a three-season pattern of menstrual cycle length across a calendar year. Overall, the difference in menstrual cycle length were modest, with the longest lengths observed from January through April, slightly shorter lengths in May through August, and the shortest lengths in September through December. Subgroup analyses suggested slightly stronger seasonality of cycle length among participant who were under age 30, who lived below 35°N and above 40°N, and who had history of PCOS, though variation was within one day and not clinically significant.

A few studies examining seasonal patterns of menstrual cycle length in different countries reported inconsistent results. One study analyzed menstrual cycle data collected among women from Minnesota and North Carolina. The main finding from Minnesota participants showed that cycles starting in November/December were approximately 0.4-0.5% longer compared to cycles starting in May/June. Results from North Carolina suggested participants who were under age 30 had a 12% longer follicular phase in early March compared to cycles starting in September³⁰. Another study in Japan found no notable seasonal differences in menstrual cycle length after controlling for environmental variables such as temperature, precipitation, and sunshine hours³¹. A study in Russia observed an approximately 0.9-day shorter menstrual cycle length in summer (June-August) compared to winter (November-January)³². A possible explanation of the inconsistent results between previous findings and our results could be the differences of geographic and climate characteristics (e.g., latitude) of the study location and residual confounding by different covariates considered across studies.

Seasonal changes of menstrual cycle patterns might be explained by various environmental and behavioral factors such as light, temperature, physical activity, and diet. In this analysis, we controlled for time-varying variables on behavioral factors including changes in exercise, weight, stressful events, sleep, alcohol use, cigarettes, marijuana, and electronic nicotine even though they are likely mediators rather than confounders as they suggest different pathways of action. In addition, adjusting for these factors did not lead to substantial changes in model estimates. Environmental light exposure could be an underlying factor for the seasonality of menstrual cycles⁷. The study of seasonality and menstrual cycle length in Russia further showed an association between longer sunshine hours in 2-3 days before ovulation and shorter cycle length after adjusting for season³². Earlier controlled exposure studies have showed light exposure was associated with shorter menstrual cycles and increased number of ovulatory cycles, and this association may be mediated by melatonin³³⁻³⁵. Melatonin secretion is responsive to environmental light and has been documented to peak in winter and trough in summer³⁶. Melatonin may suppress ovarian and androgenic activities, and higher follicular melatonin levels were associated with lower luteinizing hormone levels^{37,38}. In addition, lower levels of vitamin D are more likely in winter due to reduced ultraviolet radiation exposure from sunlight and were associated with longer menstrual cycle length³⁰. However, these seasonal changes of melatonin and vitamin D may only explain the shorter cycle length in May–August but not for the reductions observed in October–December.

Across all age subgroups, participants under age 30 showed slightly stronger seasonal trends, while little seasonal variations were found for those above age 40. Younger reproductive-aged females may be more responsive to environmental exposures than participants above age 40 whose cycles may be more influenced by diminishing ovarian reserve, masking any seasonal influence on menstrual cycles. Of note, participants who had PCOS demonstrated stronger seasonality in menstrual cycle length, but the underlying biological mechanism is not clear. Light exposure or melatonin levels might explain this, as earlier studies among PCOS patients have shown increased light exposure can decrease melatonin and lead to shorter and more regular menstrual cycles^{39,7}. However, this would only explain the shorter cycle lengths observed in summer but not in winter. Stronger seasonal variation of menstrual cycle length was also found among participants who resided in states below 35°N (e.g., Florida, Georgia, Arkansas) and above 40°N (e.g., Pennsylvania, Iowa, Oregon). Although the mechanisms were unclear, the differential estimates by latitude suggested more than one environmental (e.g., light and temperature) and/or unmeasured behavioral factor (e.g., dietary pattern) could be involved.

The timing of menstrual cycle length became shorter in our study overlapped with the timing of fecundability peak reported in other observational studies in North America (e.g., November and December)⁹. However, our findings may provide very limited support that cycle length difference can explain fecundability differences as the magnitude of changes in menstrual cycle length was small (i.e., within one day). In addition, previous studies on the associations of menstrual cycle length and fecundability reported results used binary variables of long or short cycles as the exposure (e.g., short cycles were defined as cycles shorter than 26 days and long cycles were defined as cycles longer than 29 days)²⁻⁵. Therefore, it is unclear that the small decrease in cycle length measured using a continuous variable in our study can explain the observed seasonality of fecundability.

This study has several limitations. First, measurement error and misclassification are possible as all information was self-reported. Although we controlled for several time-varying behavioral factors that may affect menstrual cycle length in the analysis, residual confounding due to unmeasured factors and/or inaccurate measures of these factors is possible. Major stressful events were not defined in the survey given this concept could vary subjectively by individual. Therefore, inconsistency in reporting major stressful events is possible and can lead to residual confounding. Data on potential underlying environmental factors that may explain the observed seasonal differences, such as temperature, light, and diet, were not available in the AWHs. Therefore, we were not able to further explore underlying biological pathways. Moreover, by excluding menstrual cycles with hormone use, participants who used hormones to manage their menstrual cycles were underrepresented in our study sample, which may have led to selection bias.

Conclusions

Overall, using data collected from mobile menstrual tracking app and surveys, our analysis demonstrated a 3-season pattern with very modest changes of menstrual cycle length among reproductive aged females in the US. In addition, larger seasonal variability was found among participants under age 30 who lived below 35°N or above 40°N and who reported a

history of PCOS. However, the observed cycle length differences were in general small and unlikely to affect clinical endpoints, such as subfertility.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Competing interests:

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Reference List

1. American Academy of Pediatrics Committee on Adolescence, American College of Obstetricians and Gynecologists Committee on Adolescent Health Care, Diaz A, Laufer MR, Breech LL. Menstruation in girls and adolescents: using the menstrual cycle as a vital sign. *Pediatrics*. 2006;118(5):2245–2250. doi: 10.1542/peds.2006-2481 [PubMed: 17079600]
2. Wesselink AK, Wise LA, Hatch EE, et al. Menstrual cycle characteristics and fecundability in a North American preconception cohort. *Ann Epidemiol*. 2016;26(7):482–487.e1. doi:10.1016/j.annepidem.2016.05.006 [PubMed: 27449569]
3. Wise LA, Mikkelsen EM, Rothman KJ, et al. A prospective cohort study of menstrual characteristics and time to pregnancy. *Am J Epidemiol*. 2011;174(6):701–709. doi:10.1093/aje/kwr130 [PubMed: 21719742]
4. Small CM, Manatunga AK, Klein M, et al. Menstrual Cycle Characteristics: Associations With Fertility and Spontaneous Abortion. *Epidemiology*. 2006;17(1):52–60. doi:10.1097/01.ede.0000190540.95748.e6 [PubMed: 16357595]
5. Jensen TK, Scheike T, Keiding N, Schaumburg I, Grandjean P. Fecundability in relation to body mass and menstrual cycle patterns. *Epidemiology*. 1999;10(4):422–428. doi:10.1097/00001648-199907000-00011 [PubMed: 10401878]
6. Mahoney MM. Shift work, jet lag, and female reproduction. *Int J Endocrinol*. 2010;2010:813764. doi:10.1155/2010/813764 [PubMed: 20224815]
7. Lee Barron M. Light Exposure, Melatonin Secretion, and Menstrual Cycle Parameters: An Integrative Review. *Biological Research For Nursing*. 2007;9(1):49–69. doi:10.1177/1099800407303337 [PubMed: 17601857]
8. Brzezinski A, Lynch HJ, Seibel MM, Deng MH, Nader TM, Wurtman RJ. The Circadian Rhythm of Plasma Melatonin During the Normal Menstrual Cycle and in Amenorrheic Women*. *The Journal of Clinical Endocrinology & Metabolism*. 1988;66(5):891–895. doi:10.1210/jcem-66-5-891 [PubMed: 3129448]
9. Wesselink AK, Wise LA, Hatch EE, et al. Seasonal patterns in fecundability in North America and Denmark: a preconception cohort study. *Human Reproduction*. 2020;35(3):565–572. doi:10.1093/humrep/dez265 [PubMed: 32003426]
10. Bronson FH. Seasonal Variation in Human Reproduction: Environmental Factors. *The Quarterly Review of Biology*. 1995;70(2):141–164. doi:10.1086/418980 [PubMed: 7610233]

11. Lam DA, Miron JA. Global patterns of seasonal variation in human fertility. *Ann N Y Acad Sci.* 1994;709:9–28. doi:10.1111/j.1749-6632.1994.tb30385.x [PubMed: 8154738]
12. Lam DA, Miron JA. The effects of temperature on human fertility. *Demography.* 1996;33(3):291–305. [PubMed: 8875063]
13. Seiver DA. Trend and variation in the seasonality of U.S. fertility, 1947-1976. *Demography.* 1985;22(1):89–100. [PubMed: 3979618]
14. Russell D, Douglas AS, Allan TM. Changing seasonality of birth--a possible environmental effect. *J Epidemiol Community Health.* 1993;47(5):362–367. doi:10.1136/jech.47.5.362 [PubMed: 8289035]
15. Haus E, Lakatua DJ, Halberg F, et al. Chronobiological Studies of Plasma Prolactin in Women in Kyushu, Japan, and Minnesota, USA. *The Journal of Clinical Endocrinology & Metabolism.* 1980;51(3):632–640. doi:10.1210/jcem-51-3-632 [PubMed: 7410538]
16. Garde AH, Hansen ÅM, Skovgaard LT, Christensen JM. Seasonal and Biological Variation of Blood Concentrations of Total Cholesterol, Dehydroepiandrosterone Sulfate, Hemoglobin A1c, IgA, Prolactin, and Free Testosterone in Healthy Women. *Clinical Chemistry.* 2000;46(4):551–559. doi:10.1093/clinchem/46.4.551 [PubMed: 10759480]
17. Kauppila A, Pakarinen A, Kirkinen P, Mäkilä U. The effect of season on the circulating concentrations of anterior pituitary, ovarian and adrenal cortex hormones and hormone binding proteins in the subarctic area; evidence of increased activity of the pituitary-ovarian axis in spring. *Gynecol Endocrinol.* 1987;1(2):137–150. doi:10.3109/09513598709030678 [PubMed: 3140579]
18. Martikainen H, Ruokonen A, Tomás C, Kauppila A. Seasonal changes in pituitary function: amplification of midfollicular luteinizing hormone secretion during the dark season. *Fertil Steril.* 1996;65(4):718–720. doi:10.1016/s0015-0282(16)58202-8 [PubMed: 8654627]
19. Mahalingaiah S, Fruh V, Rodriguez E, et al. Design and methods of the Apple Women's Health Study: a digital longitudinal cohort study. *American Journal of Obstetrics and Gynecology.* Published online October 2021 :S0002937821010929. doi:10.1016/j.ajog.2021.09.041
20. Li H, Gibson EA, Jukic AMZ, et al. Menstrual cycle length variation by demographic characteristics from the Apple Women's Health Study, *npj Digit Med.* 2023;6(1):100. doi:10.1038/s41746-023-00848-1 [PubMed: 37248288]
21. Bull JR, Rowland SP, Scherwitzl EB, Scherwitzl R, Danielsson KG, Harper J. Real-world menstrual cycle characteristics of more than 600,000 menstrual cycles, *npj Digit Med.* 2019;2(1):83. doi:10.1038/s41746-019-0152-7 [PubMed: 31482137]
22. Campbell LR, Scalise AL, DiBenedictis BT, Mahalingaiah S. Menstrual cycle length and modern living: a review. *Curr Opin Endocrinol Diabetes Obes.* 2021;28(6):566–573. doi: 10.1097/MED.0000000000000681 [PubMed: 34545843]
23. Grieger JA, Norman RJ. Menstrual Cycle Length and Patterns in a Global Cohort of Women Using a Mobile Phone App: Retrospective Cohort Study. *J Med Internet Res.* 2020;22(6):e17109. doi:10.2196/17109 [PubMed: 32442161]
24. Harlow SD, Ephross SA. Epidemiology of menstruation and its relevance to women's health. *Epidemiol Rev.* 1995;17(2):265–286. doi:10.1093/oxfordjournals.epirev.a036193 [PubMed: 8654511]
25. Adler N, Singh-Manoux A, Schwartz J, Stewart J, Matthews K, Marmot MG. Social status and health: a comparison of British civil servants in Whitehall-II with European- and African-Americans in CARDIA. *Soc Sci Med.* 2008;66(5):1034–1045. doi:10.1016/j.socscimed.2007.11.031 [PubMed: 18180089]
26. Stolwijk AM, Straatman H, Zielhuis GA. Studying seasonality by using sine and cosine functions in regression analysis. *J Epidemiol Community Health.* 1999;53(4):235–238. doi:10.1136/jech.53.4.235 [PubMed: 10396550]
27. Munro MG, Critchley HOD, Fraser IS, the FIGO Menstrual Disorders Committee. The two FIGO systems for normal and abnormal uterine bleeding symptoms and classification of causes of abnormal uterine bleeding in the reproductive years: 2018 revisions. *Int J Gynecol Obstet.* 2018;143(3):393–408. doi:10.1002/ijgo.12666

28. Gibson EA, Li H, Fruh V, et al. Covid-19 vaccination and menstrual cycle length in the Apple Women's Health Study, npj Digit Med. 2022;5(1):165. doi:10.1038/s41746-022-00711-9 [PubMed: 36323769]
29. Lum KJ, Sundaram R, Louis TA. Accounting for length-bias and selection effects in estimating the distribution of menstrual cycle length. Biostatistics. 2015;16(1):113–128. doi:10.1093/biostatistics/kxu035 [PubMed: 25027273]
30. Jukic AMZ, Wilcox AJ, McConaughey DR, Weinberg CR, Steiner AZ. Lower 25-Hydroxyvitamin D and Long Menstrual Cycles in a Prospective Cohort Study. Epidemiology. 2018;29(3):388–396. doi:10.1097/EDE.0000000000000804 [PubMed: 29337846]
31. Tatsumi T, Sampei M, Saito K, et al. Age-Dependent and Seasonal Changes in Menstrual Cycle Length and Body Temperature Based on Big Data. Obstet Gynecol. 2020;136(4):666–674. doi:10.1097/AQG.0000000000003910 [PubMed: 32925608]
32. Danilenko KV, Sergeeva OY, Verevkin EG. Menstrual cycles are influenced by sunshine. Gynecol Endocrinol. 2011;27(9):711–716. doi:10.3109/09513590.2010.521266 [PubMed: 20937003]
33. Dewan EM. On the possibility of a perfect rhythm method of birth control by periodic light stimulation. American Journal of Obstetrics and Gynecology. 1967;99(7):1016–1019. doi:10.1016/0002-9378(67)90260-8 [PubMed: 6058735]
34. Dewan EM, Menkin MF, Rock J. EFFECT OF PHOTIC STIMULATION ON THE HUMAN MENSTRUAL CYCLE. Photochem Photobiol. 1978;27(5):581–585. doi:10.1111/j.1751-1097.1978.tb07649.x [PubMed: 674392]
35. Danilenko KV, Samoilova EA. Stimulatory effect of morning bright light on reproductive hormones and ovulation: results of a controlled crossover trial. PLoS Clin Trials. 2007;2(2):e7. doi:10.1371/journal.pctr.0020007 [PubMed: 17290302]
36. Wehr TA. Melatonin and Seasonal Rhythms. J Biol Rhythms. 1997;12(6):518–527. doi:10.1177/074873049701200605 [PubMed: 9406025]
37. Kauppila A, Kivelä A, Pakarinen A, Vakkuri O. Inverse Seasonal Relationship Between Melatonin and Ovarian Activity in Humans in a Region With a Strong Seasonal Contrast in Luminosity*. The Journal of Clinical Endocrinology & Metabolism. 1987;65(5):823–828. doi:10.1210/jcem-65-5-823 [PubMed: 3667880]
38. Kivelä A, Kauppila A, Ylöstalo P, Vakkuri O, Leppäluoto J. Seasonal, menstrual and circadian secretions of melatonin, gonadotropins and prolactin in women. Acta Physiologica Scandinavica. 1988;132(3):321–327. doi:10.1111/j.1748-1716.1988.tb08335.x [PubMed: 3147572]
39. Luboshitzky R, Qupti G, Ishay A, Shen-Orr Z, Futerman B, Linn S. Increased 6-sulfatoxymelatonin excretion in women with polycystic ovary syndrome. Fertility and Sterility. 2001;76(3):506–510. doi:10.1016/S0015-0282(01)01930-6 [PubMed: 11532473]

Highlights

- Digital cohort of 125,104 menstrual cycles from 17,427 participants within the US
- Modest seasonal variation of menstrual cycle length from June 2020 to July 2022
- Shorter cycles (<1/5 day) in May-Aug and Sept-Dec compared to Jan-April
- Younger participants <35°N and >40°N with PCOS had strongest seasonal trends
- No seasonal patterns in cycle lengths were found for participants above age 40

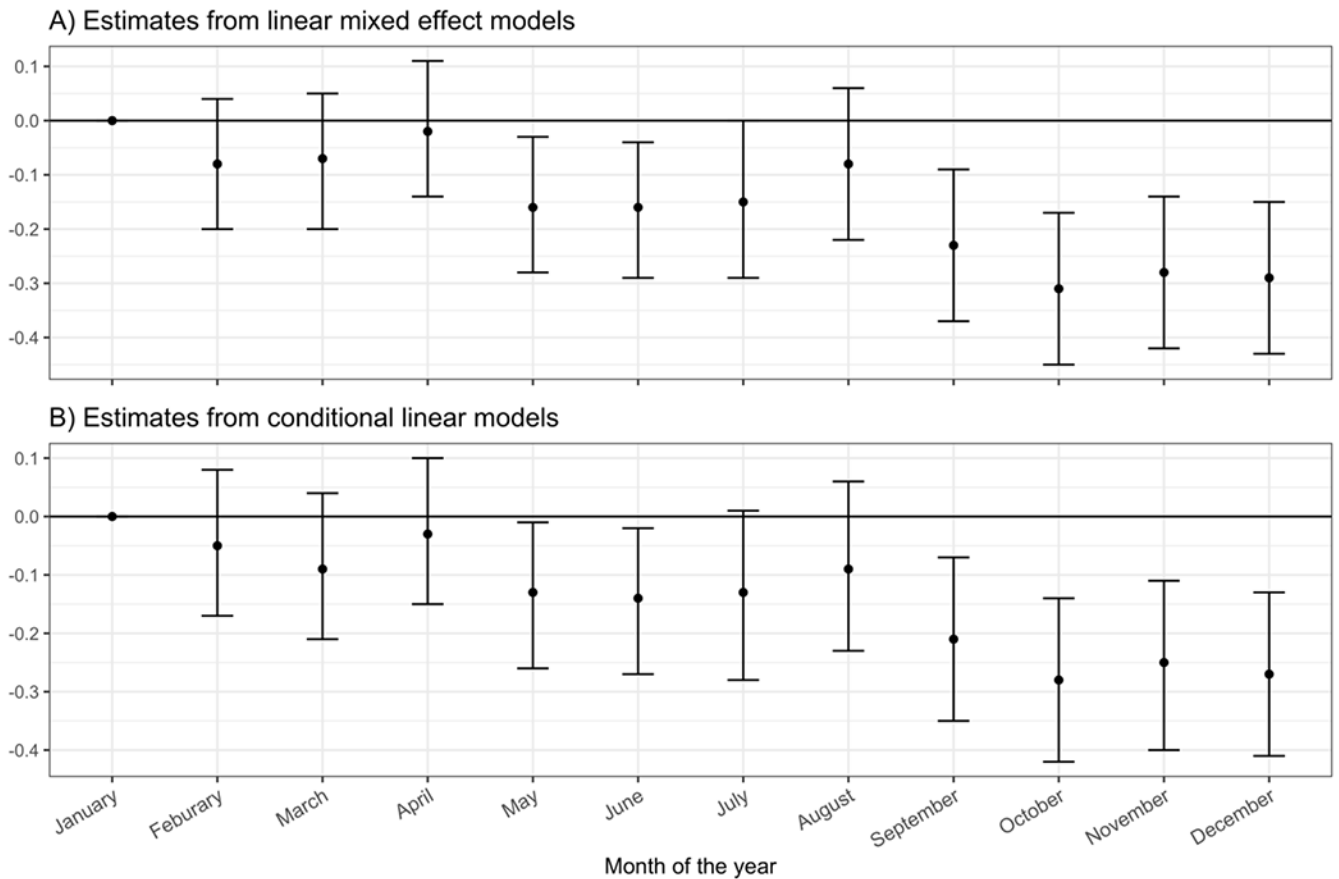


Figure 1.

Adjusted differences of menstrual cycle length (days) and 95% confidence intervals by calendar months from (A) linear mixed effect models and (B) conditional linear models among 125104 menstrual cycles from 17427 participants of the Apple Women's Health Study

Table 1

Distribution of demographic and lifestyle factors associated with menstrual cycle length in 125104 menstrual cycles from 17427 participants of the Apple Women's Health Study

		N (%)
N of menstrual cycles		125104
Age		
	Under 25	15464 (12.4)
	25-29	18198 (14.5)
	30-34	23993 (19.2)
	35-39	27180 (21.7)
	40-45	24936 (19.9)
	45-49	15333 (12.3)
Race and ethnicity		
	White, non-Hispanic	91432 (73.1)
	Black, African American, or African	6285 (5.0)
	Asian	4988 (4.0)
	Hispanic	8157 (6.5)
	Other	2539 (2.0)
	More than one	11420 (9.1)
	Missing	283 (0.2)
BMI ^a		
	Underweight	2670 (2.1)
	Healthy	43091 (34.4)
	Overweight	32006 (25.6)
	Obese	45489 (36.4)
	Missing	1848 (1.5)
Education		
	High school and less	12850 (10.3)
	3-year college or technical schools	35341 (28.2)
	4-year college	42325 (33.8)
	Graduate school	34306 (27.4)
	Missing	282 (0.2)
MacArthur scale of subjective social status		
	Low (0-3)	27257 (21.8)
	Medium (4-6)	77518 (62)
	High (7-9)	20225 (16.2)
	Missing	104 (0.1)
Change in exercise		
	Decrease	15676 (12.5)
	No change	96781 (77.4)
	Increase	12124 (9.7)
	Missing	523 (0.4)

		N (%)
Change in weight		
	Decrease	3647 (2.9)
	No change	115191 (92.1)
	Increase	5817 (4.6)
	Missing	449 (0.4)
Change in health conditions ^b		
	No	108936 (87.1)
	Yes	15877 (12.7)
	Missing	291 (0.2)
Sleep disruption		
	No	105110 (84.0)
	Yes	19703 (15.7)
	Missing	291 (0.2)
Major stressful event		
	No	100705 (80.5)
	Yes	24108 (19.3)
	Missing	291 (0.2)
Alcohol use		
	No drinking	41141 (32.9)
	1-5 per month	39073 (31.2)
	6-10 per month	15582 (12.5)
	11-15 per month	10405 (8.3)
	16+ per month	14376 (11.5)
	Missing	4527 (3.6)
Cigarette smoking		
	No	110270 (88.1)
	Yes	10392 (8.3)
	Missing	4442 (3.6)
Marijuana use		
	No	93246 (74.5)
	Yes	26954 (21.5)
	Missing	4904 (3.9)
E-cigarette use		
	No	110402 (88.2)
	Yes	10336 (8.3)
	Missing	4366 (3.5)
History of PCOS		
	No	112263 (89.7)
	Yes	12525 (10.0)
	Missing	316 (0.3)
Calendar year		
	2020	21325 (17.0)

	N (%)	
	2021	68956 (55.1)
	2022	34823 (27.8)
Latitude ^c		
	<35° N	29789 (23.8)
	35°-39° N	44864 (35.9)
	≥40° N	50137 (40.1)
	US territories	314 (0.3)

Abbreviations: BMI, body mass index; PCOS, polycystic ovary syndrome.

^aUnderweight: BMI<18.5 kg/m²; Healthy: 18.5≤BMI<25 kg/m²; Overweight: 25≤BMI<30 kg/m²; Obese: BMI≥30 kg/m².

^bIncluding change of medication, hospitalization, had significant illness, and had surgery.

^cLatitude was measured by the latitude of the centroid of the state of residence reported by the participants. States below 35° N include: Florida, Georgia, South Carolina, Alabama, Mississippi, Louisiana, Texas, Arkansas, New Mexico, Arizona, and Hawaii; between 35°-39° N: Indiana, Missouri, Kansas, West Virginia, Maryland, North Carolina, Delaware, Virginia, District of Columbia, Tennessee, Kentucky, Oklahoma, Utah, Colorado, Nevada, and California; above 40° N: Rhode Island, New Hampshire, Vermont, Connecticut, Massachusetts, New Jersey, Pennsylvania, New York, Illinois, Wisconsin, Ohio, Michigan, Nebraska, South Dakota, Idaho, Iowa, Wyoming, Oregon, Maine, Minnesota, North Dakota, Montana, Washington, and Alaska.

Table 2

Seasonality of menstrual cycle length estimated from sinusoidal model among 125104 menstrual cycles from 17427 participants of the Apple Women’s Health Study

	Coefficients		Differences of MCL between the longest and shortest months (days)	Null distribution mean (P2.5, P97.5)	Months with the longest and shortest cycles	
	sine	cosine			Longest	Shortest
Crude ^a	0.08	-0.02	0.16 (0.06, 0.26)	0.05 (0.01, 0.11)	April	October
Parsimonious ^b	0.08	-0.03	0.16 (0.06, 0.26)	0.05 (0.01, 0.11)	April	October
Fully adjusted ^c	0.07	-0.03	0.16 (0.06, 0.26)	0.05 (0.01, 0.11)	April	October

^a Only adjusting for calendar year

^b Adjusting for calendar year, age, body mass index, race and ethnicity, MacArthur Social Stress Scale, and education

^c Adjusting for calendar year, age, body mass index, race and ethnicity, MacArthur Social Stress Scale, education, changes of exercise, changes of weight, major health changes (i.e., change of medication, hospitalization, had significant illness, and had surgery), major stressful event, sleep disruption, smoking, alcohol use, marijuana use, and e-cigarette use.

Table 3

Adjusted cycle length differences (days) and 95% confidence intervals by calendar months from linear mixed effect models stratified by age among 125104 menstrual cycles from 17427 participants of the Apple Women's Health Study

	Under 25	25-29	30-34	35-39	Above 40
N cycles (N participants)	15464 (2690)	18198 (2885)	23993 (3579)	27180 (3534)	40269 (4739)
January	Reference	Reference	Reference	Reference	Reference
February	-0.23 (-0.60, 0.15)	-0.12 (-0.45, 0.22)	-0.13 (-0.39, 0.13)	-0.06 (-0.30, 0.18)	0.03 (-0.20, 0.26)
March	-0.12 (-0.50, 0.26)	-0.01 (-0.34, 0.33)	-0.13 (-0.39, 0.12)	-0.04 (-0.28, 0.19)	-0.07 (-0.30, 0.16)
April	-0.01 (-0.40, 0.38)	-0.04 (-0.38, 0.03)	-0.10 (-0.37, 0.17)	-0.15 (-0.39, 0.09)	0.17 (-0.07, 0.40)
May	-0.33 (-0.72, 0.05)	-0.10 (-0.45, 0.24)	-0.32 (-0.58, -0.06)	-0.27 (0.51, -0.03)	0.08 (-0.15, 0.31)
June	-0.07 (-0.46, 0.31)	-0.31 (-0.65, 0.03)	-0.29 (-0.56, -0.03)	-0.25 (-0.49, -0.02)	0.01 (-0.22, 0.24)
July	-0.31 (-0.75, 0.13)	-0.13 (-0.52, 0.26)	-0.06 (-0.36, 0.24)	-0.29 (-0.57, -0.02)	-0.04 (-0.31, 0.22)
August	-0.33 (-0.77, 0.11)	0.01 (-0.40, 0.40)	-0.17 (-0.47, 0.13)	0.03 (-0.25, 0.31)	-0.02 (-0.28, 0.24)
September	-0.59 (-1.02, -0.15)	-0.19 (-0.58, 0.20)	-0.17 (-0.48, 0.13)	-0.37 (0.65, -0.09)	-0.02 (-0.28, 0.24)
October	-0.76 (-1.19, -0.33)	-0.46 (0.84, -0.07)	-0.30 (-0.60, 0.00)	-0.40 (-0.67, -0.13)	0.03 (-0.23, 0.29)
November	-0.44 (-0.87, -0.01)	-0.57 (-0.95, -0.19)	-0.19 (-0.49, 0.11)	-0.37 (-0.65, -0.10)	-0.03 (-0.29, 0.24)
December	-0.41 (-0.83, 0.01)	-0.51 (-0.88, -0.13)	-0.44 (-0.74, -0.15)	-0.34 (-0.61, -0.07)	-0.01 (-0.26, 0.25)

Adjusting for calendar year, age, body mass index, race and ethnicity, MacArthur Social Stress Scale, education, changes of exercise, changes of weight, major health changes (i.e., change of medication, hospitalization, had significant illness, and had surgery), major stressful event, sleep disruption, smoking, alcohol use, marijuana use, e-cigarette use, and a random intercept by participant

Table 4

Adjusted cycle length differences (days) and 95% confidence intervals by calendar months from linear mixed effect models stratified by latitude and polycystic ovary syndrome (PCOS) diagnosis among participants of the Apple Women’s Health Study

N cycles (N participants)	By latitude ^a			By PCOS status		
	<35° N	35°-39° N	≥40° N	PCOS	No PCOS	
	Reference	Reference	Reference	Reference	Reference	
January	29789 (4351)	44864 (6310)	50137 (6730)	12525 (2003)	112263 (15388)	
February	-0.08 (-0.34, 0.17)	0.04 (-0.17, 0.24)	-0.20 (-0.42, 0.01)	-0.11 (-0.63, 0.41)	-0.08 (-0.2, 0.05)	
March	-0.07 (-0.33, 0.18)	0.00 (-0.21, 0.20)	-0.14 (-0.36, 0.07)	-0.15 (-0.67, 0.37)	-0.07 (-0.2, 0.05)	
April	-0.18 (-0.44, 0.08)	0.12 (-0.09, 0.33)	-0.06 (-0.28, 0.15)	0.11 (-0.41, 0.63)	-0.04 (-0.17, 0.09)	
May	-0.20 (-0.46, 0.06)	-0.01 (-0.22, 0.19)	-0.31 (-0.52, -0.09)	-0.05 (-0.57, 0.47)	-0.18 (-0.30, -0.05)	
June	-0.17 (-0.43, 0.09)	-0.15 (-0.36, 0.05)	-0.23 (-0.44, -0.02)	-0.76 (-1.28, -0.24)	-0.10 (-0.23, 0.02)	
July	-0.12 (-0.42, 0.18)	0.07 (-0.17, 0.31)	-0.48 (-0.73, -0.23)	-0.38 (-0.98, 0.22)	-0.13 (-0.27, 0.02)	
August	-0.35 (-0.64, -0.05)	0.09 (-0.15, 0.33)	-0.12 (-0.37, 0.12)	0.14 (-0.46, 0.73)	-0.10 (-0.25, 0.04)	
September	-0.20 (-0.49, 0.10)	-0.04 (-0.28, 0.20)	-0.50 (-0.75, -0.25)	-0.54 (-1.13, 0.05)	-0.21 (-0.35, -0.07)	
October	-0.31 (-0.60, -0.02)	-0.12 (-0.04, 0.11)	-0.52 (-0.76, -0.28)	-0.67 (-1.26, -0.09)	-0.28 (-0.42, -0.14)	
November	-0.40 (-0.69, -0.10)	-0.19 (-0.42, 0.05)	-0.31 (-0.56, -0.07)	-0.59 (-1.19, 0.00)	-0.25 (-0.39, -0.10)	
December	-0.42 (-0.71, -0.13)	-0.19 (-0.42, 0.04)	-0.36 (-0.60, -0.13)	-0.44 (-1.02, 0.14)	-0.29 (-0.43, -0.15)	

Adjusting for calendar year, age, body mass index, race and ethnicity, MacArthur Social Stress Scale, education, changes of exercise, changes of weight, major health changes (i.e., change of medication, hospitalization, had significant illness, and had surgery), major stressful event, sleep disruption, smoking, alcohol use, marijuana use, and a random intercept by participants.

^aLatitude was measured by the latitude of the centroid of the state of residence reported by the participants. States below 35° N include: Florida, Georgia, South Carolina, Alabama, Mississippi, Louisiana, Texas, Arkansas, New Mexico, Arizona, and Hawaii; between 35°-39° N: Indiana, Missouri, Kansas, West Virginia, Maryland, North Carolina, Delaware, Virginia, District of Columbia, Tennessee, Kentucky, Oklahoma, Utah, Colorado, Nevada, and California; above 40° N: Rhode Island, New Hampshire, Vermont, Connecticut, Massachusetts, New Jersey, Pennsylvania, New York, Illinois, Wisconsin, Ohio, Michigan, Nebraska, South Dakota, Idaho, Iowa, Wyoming, Oregon, Maine, Minnesota, North Dakota, Montana, Washington, and Alaska. Participants who reported residing in US territories were excluded.